ISL SIGN RECOGNITION

Model and project synopsis

A detailed guide to the dataset, training process, MediaPipe landmark training, live inference, text controls, evaluation, and known limits.

76.38% with hand-count filter	**2,430 detected of 3,370 test images** 72.11% landmark coverage	**26 letters, A-Z**

What this project does

This Linux desktop prototype recognizes a **single still alphabet sign** at a time and can place confirmed letters into an on-screen text buffer. It is a fingerspelling aid and a learning prototype. It does not recognize connected signing, translate sentences, or understand sign-language grammar. Letter-by-letter output is not the same thing as full ISL translation.

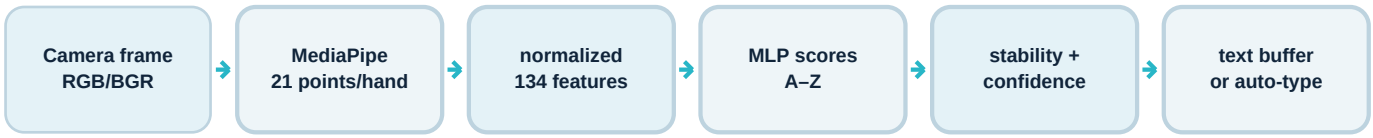
Current build at a glance

Area	Current implementation
Input	Webcam video; MediaPipe Hands returns up to two sets of 21 hand landmarks.
Classifier	A compact 26-class MLP trained directly on normalized hand landmark features.
Desktop UI	Native Tkinter window; camera preview, current letter, confidence, text and controls.
Deployment weights	artifacts/static_model.pt (loaded by the webcam app).
Evaluation	76.38% accuracy after hand-count filter; 72.11% coverage (2,430/3,370).

Prepared from the project source and its recorded training/evaluation artifacts. The exact score reflects this saved checkpoint and split; it should not be read as expected accuracy for a new person or webcam setup.

1. System overview

The application has two connected jobs: turn a camera frame into a candidate letter, then decide whether that candidate is stable enough to show or type. MediaPipe provides the hand geometry; a learned landmark model chooses the letter.



The runtime loop in plain language

- **Read a frame.** The webcam gives the app a color image. The classifier path now keeps the same left/right orientation used by the training photos; the displayed preview is mirrored for a natural mirror-like view.
- **Locate hands.** MediaPipe Hands estimates 21 points per detected hand. Up to two hands are ordered left-to-right. If no hand is detected, the app does not send a guess.
- **Prepare the geometry.** For each hand, the 21 x/y/z points are wrist-relative and normalized by hand span. The vector also records normalized hand center, scale and presence. Two hands form 134 input values; an absent second hand is represented by zeros.
- **Score the letters.** The MLP returns 26 scores. Softmax turns these into normalized scores. The highest score is the leading guess; the app rejects it if confidence or the gap from second place is too small.
- **Stabilize.** Recent accepted guesses are counted across frames. A letter appears as confirmed only after enough frames agree.
- **Show or type.** The confirmed letter can be manually added, or auto-type can append it after a 500 ms hold. Duplicate prevention waits for a hand to leave or a different sign before typing again.

What the classifier sees

The current classifier is a landmark MLP: its input is the normalized MediaPipe joint vector, not image pixels. The training pipeline extracts the same 134-value representation from labeled static images, caches it, and trains from the existing duplicate-aware split.

2. Dataset and split

The project uses the ISLRTC-referred image collection attributed to Atharva Dumbre. Images are stored in class folders. The static recognition task includes every English alphabet class A–Z, including J and K. Digit folders are not part of this model.

The saved manifest contains the following partition sizes:

Partition	Images	Purpose
Train	18,533	Used to update landmark MLP weights for images with detected hands.
Validation	4,096	Used during training to track generalization and choose the saved best checkpoint.
Test	3,370	Held aside for final evaluation; not used for gradient updates.
Total	25,999	26 static letter classes.

How the split is made

Before assigning images to partitions, the preparation code computes a perceptual difference hash within each letter class. Images with hashes within Hamming distance 3 are grouped as likely exact or near duplicates. Whole groups are assigned to one partition so close copies are less likely to leak between train, validation and test. The intended proportions are approximately 70% / 15% / 15% per class, with rounding and grouping affecting exact totals.

What this protects against

It reduces evaluation inflation from repeated or visually near-identical images appearing on both sides of the split. It does not establish signer independence: the source dataset does not provide verified signer or capture-session IDs. Therefore, images from the same person or session may still occur across partitions.

Data rights note

Project metadata describes the Kaggle source as CC BY-NC-SA 4.0. Review the source terms and attribution/share-alike requirements before redistributing source images or derived model files. The dataset is used here as research/prototype input; no independent legal audit is claimed.

Pipeline stages

Command	Role
isl-validate-static	Checks expected class folders and image readability; reports duplicates/conflicts.
isl-prepare-static	Builds the manifest, duplicate groups and train/validation/test split.
isl-train-static	Extracts MediaPipe features, then trains the static A–Z landmark MLP.
isl-evaluate-static	Scores the held-out test partition and writes the report/confusion matrix.

3. Landmark model and training

The classifier is a compact multilayer perceptron (MLP) implemented in PyTorch. It starts from random weights and consumes normalized MediaPipe joint geometry. It learns combinations of finger-joint positions, finger bends, hand orientation, hand size and the relative arrangement of up to two hands.

Network shape

Stage	Operation	Output / role
Input	Two hand slots; 21 xyz points per hand	134 normalized values
Hidden 1	Linear, layer norm, ReLU, dropout 0.30	256 features
Hidden 2	Linear, layer norm, ReLU, dropout 0.20	128 features
Hidden 3	Linear, ReLU	64 features
Output	Linear classifier	26 class logits (A–Z)

Training loop

The pipeline runs MediaPipe Hands on each labeled still image, upscaling smaller images for landmark detection. It caches detected feature vectors in artifacts/static_landmark_features.npz. Images where no hand is found are omitted from fitting and counted in the training output. Remaining vectors retain their train/validation/test assignment. Training uses class-weighted cross-entropy and AdamW (learning rate 0.001, weight decay 0.0001). Seed 42 is recorded for repeatability; validation loss drives early stopping.

Checkpoint behavior

At each epoch the project writes a resumable last checkpoint and, when validation loss improves, a best checkpoint. After training, it exports the deployment model at **artifacts/static_model.pt**. The webcam and evaluator reconstruct the MLP from architecture and feature metadata in that checkpoint. The saved best decision is based on validation loss, not the highest validation accuracy.

Recorded run

Recorded item	Value
Epochs in saved history	17
Highest validation accuracy	83.78% at epoch 15
Lowest validation loss	0.6794 at epoch 11
Test accuracy	78.48% among 2,430 images with detected hands
Landmark coverage	72.11% (2,430/3,370)
Coverage-adjusted accuracy	56.59% counting missed hands as failures
Test evaluation latency	0.17 ms per image, model-only on cpu

The latency figure measures only the model forward pass in the saved report. It excludes camera capture, MediaPipe hand tracking and UI drawing. It is not the end-to-end frame rate.

4. Webcam application and text controls

The native desktop interface is built with Tkinter. OpenCV supplies camera frames, MediaPipe Hands provides live landmarks, and PyTorch runs the MLP on normalized joint features. The app uses a single camera loop and refreshes the preview while updating the current guess and text panel.

Recognition gates

Gate	What it does	What it does not do
Background calibration	Samples the empty camera view at startup; a frame-difference check estimates whether a foreground object is present.	It is not a hand detector or image segmentation model. Lighting changes and movement can affect it.
MediaPipe Hands	Detects up to two hands and returns 21 normalized landmarks per detected hand. Points feed the MLP and appear on the preview.	It can miss occluded, small, or poorly lit hands. If it misses, the app withholds a letter.
MLP confidence/margin	Rejects uncertain guesses when top score is below the confidence threshold or too close to the second score.	Softmax scores are not calibrated real-world probabilities. High confidence can still be wrong.
Frame confirmation	Requires agreement across recent frame guesses before displaying a confirmed sign.	Temporal agreement cannot repair a systematic class bias.

Controls

ADD LETTER / A: append the confirmed letter. **SPACE:** add a word space. **BACKSPACE:** delete one character. **CLEAR / C:** clear the text. **AUTO-TYPE / T:** enable or disable typing. **P:** pause/resume. **F:** fullscreen. **Q or Escape:** quit. Auto-type is off at launch; with it enabled, a stable sign must remain for 500 ms. Duplication prevention avoids repeatedly entering a held letter; the hand must leave or a new sign must be confirmed.

Launch paths

Use **Launch ISL Webcam.desktop** or **launch-isl-webcam.sh** to open the desktop app. The optional menu script at **ui startup/isl-launcher.sh** offers training and webcam choices. The command-line entry point is **isl-webcam --camera 0**. The app loads **artifacts/static_model.pt** by default.

The webcam needs a desktop session and a working camera. Leave the view empty during the initial calibration prompt, then place the hands clearly in frame.

5. Evaluation and what the score means

The held-out report gives **78.48% accuracy among 2,430 test images where MediaPipe found hands**. MediaPipe found hands in 3,370 source test images (72.11% coverage). If each missed detection is counted as an end-to-end failure, the combined score is 56.59%. The conditional accuracy measures letter classification after successful detection; coverage measures whether features are available. Neither number guarantees similar results on a new signer or webcam setup.

Some per-letter recall values in the saved report

Letter	Recall	Letter	Recall
J	24%	G	100%
I	27%	K	100%
N	34%	P	100%
M	51%	T	100%
F	52%	W	100%
Q	54%	—	0%
Y	72%	—	0%
U	73%	—	0%

Recall is the fraction of true examples of that letter that the model found. Class performance varies substantially; the lowest-recall letters in this report are S, I, T, J, Z, N, K and M. This pattern is consistent with the model needing more robust examples and/or better separation for similar handshapes.

Why the app can still misread a live sign

- The recorded test images and a webcam image differ in signer, camera distance, lighting, background, skin tone, framing, orientation and image quality.
- MediaPipe can miss hands or estimate finger joints inaccurately, especially with occlusion, motion blur, small hands or unusual camera angles.
- A static image model cannot distinguish signs that require motion over time; J and Z have movement components in common alphabet conventions. This build treats the source dataset examples as static class images, not motion sequences.
- Frame smoothing, a confidence threshold, and auto-type dwell reduce flicker and accidental repeats. They do not increase the underlying model's recognition accuracy.
- The source has no verified signer/session IDs. Near-duplicate grouping helps, but the held-out score may still benefit from shared capture conditions or signers.

Improvement priorities

For a stronger real-use system, collect labeled webcam examples from intended users and conditions, improve landmark coverage for weak classes, and evaluate on a signer-separated set. Keep confusion matrices and per-class results in the loop; a single overall accuracy can hide classes that rarely work. Never interpret a label as certain solely because its displayed confidence is high.

6. Reproducing the project

The repository includes a Python 3.12 environment, project commands, dataset manifest, model artifacts, and generated reports. A typical clean setup and run uses the commands below. The data are already present in the current workspace.

```
cd /home/advik/ISL_LIVE/isl-linux
source .venv/bin/activate
isl-validate-static
isl-prepare-static
isl-train-static --epochs 25 --batch-size 32
isl-evaluate-static
isl-webcam --camera 0
```

The first training run extracts and caches MediaPipe features; later runs reuse the cache. The trainer saves checkpoints and curves under **artifacts/**. Evaluation writes its per-class report and confusion matrix under **reports/generated/**. If resuming a run, pass the matching saved last checkpoint with `--resume`; resume metadata such as labels, seed, data split and counts must agree.

Project file map

Location	Purpose
src/isl_recognition/models/landmark_mlp.py	134-input landmark MLP and 26-class output head.
src/isl_recognition/inference/landmarks.py	Two-hand feature encoding and MediaPipe feature extraction.
src/isl_recognition/data/landmark_datasets.py	Feature-cache construction and train/validation/test dataset wiring.
src/isl_recognition/data/prepare.py	Near-duplicate grouping and split manifest creation.
src/isl_recognition/training/train_static.py	Dataset wiring and static model training entry point.
src/isl_recognition/training/common.py	Shared training loop, checkpoint handling, early stopping and curves.
src/isl_recognition/app/native_ui.py	Desktop controls, live landmark inference, confidence/stability and auto-type.
src/isl_recognition/app/webcam.py	Prediction helper, camera/model preprocessing and 500 ms dwell duplicate guard.
src/isl_recognition/inference/background.py	Startup background calibration and foreground difference gate.
artifacts/static_model.pt	Compact checkpoint loaded by the running app.
reports/generated/static_test_report.json	Saved held-out test metrics and per-letter results.

Troubleshooting quick guide

Symptom	Check
Camera unavailable	Camera permissions/index; run from a graphical desktop session; close other apps using the camera.
No stable sign	Ensure calibration was done on an empty view; keep hands fully visible; check lighting and foreground movement.
One letter repeats incorrectly	The UI may be stable while the classifier is biased. Check raw class confidence and test/live examples; smoothing cannot fix a wrong model.
Model missing	Confirm artifacts/static_model.pt exists; train using the project command.
Frequent repeated letters	Auto-type only repeats after a leave/new-sign reset; hold the new sign clearly and verify the hand drops out between letters when intended.

Summary

This is an end-to-end static alphabet recognition prototype: labeled image folders become a grouped split; MediaPipe converts images into hand-joint features; an MLP learns 26 static letters; the webcam produces the same features live; the native UI applies confidence and temporal checks; and a text buffer supports manual or dwell-based letter entry. The main evidence-backed limitation is generalization: held-out accuracy and per-letter recall are shown above, but the source split is not signer-independent. Treat output as a best-effort suggestion that needs user confirmation.

Dataset source: Kaggle, “Indian Sign Language (ISLRTC referred),” Atharva Dumbre, version 1. Project metadata records CC BY-NC-SA 4.0. Project facts and metrics above come from the checked-in manifest, saved training history, evaluation report, and implementation.

Model update: hand-count-aware inference

This update describes the current checkpoint and inference behavior. The updated test results below supersede the earlier overall accuracy and coverage-adjusted figures in the original synopsis.

Why the change was needed

The landmark model receives up to two hand point sets. Previously, it always chose among all 26 letters, even when MediaPipe detected only one hand. If a two-hand sign was partly occluded or one hand was missed, the model could still return a letter that usually appears with two detected hands.

The hand-count rule

The camera loop already knows whether MediaPipe returned one or two hands. Training now saves each letter's one-hand and two-hand sample counts in the model checkpoint. After the MLP produces its letter scores, inference keeps a letter eligible only if at least 25% of that letter's detected training examples had the same number of hands as the current frame. Ineligible letters have their score suppressed; the remaining scores are normalized before the usual confidence, margin and frame-stability checks.

With one hand visible, the app therefore selects only among letters with meaningful one-hand support in the training data. With two hands visible, it uses the corresponding two-hand support. In the current training split, 19 of 26 classes meet the support threshold for each hand count. The eligible groups overlap but are not identical.

Where it is applied

The same rule is used by live webcam recognition, single-image prediction and held-out evaluation. The MLP architecture and 134-value landmark input stay the same. The model does not run a second hand-count network: MediaPipe supplies the count, and checkpoint metadata supplies the class support.

This is a practical filter based on the dataset, not a guarantee that every sign has one fixed hand count. If MediaPipe misses a hand, a two-hand sign can be treated as one-hand input. A class with few one-hand samples may also be excluded even when a person can naturally produce that sign with one hand.

Current model files

Deployment checkpoint: artifacts/static_model.pt. The checkpoint must include per-class training_hand_count_counts metadata for the constraint to apply. Older checkpoints without this metadata retain unrestricted 26-class predictions.

Updated evaluation and interpretation

Held-out test results after applying the hand-count rule

Measure	Updated result	How to read it
Accuracy when hands are detected	**76.38%** (2,430 images)	Letter accuracy on test images where MediaPipe returned landmarks.
Hand detection coverage	**72.11%** (2,430 of 3,370)	Share of source test images with detected hands; this is unchanged by score filtering.
Coverage-adjusted accuracy	**55.07%**	Counts images with no detected hands as failures as well as classification errors.
Model forward-pass latency	**0.16 ms** (CPU)	MLP-only timing; excludes camera, MediaPipe and UI work.

What changed in the score

The unfiltered classifier scored 78.48% on detected test images. With the new count constraint, the same 2,430 detected test images score 76.38%, a 2.10 percentage-point decrease. This tradeoff is expected: the filter removes predictions that conflict with hand-count support, but can also remove the correct class when the training examples or MediaPipe count are inconsistent.

The landmark model still has meaningful recognition limits

The test partition is drawn from the same image collection and is not signer-independent. MediaPipe misses hands in about 28% of the held-out images. Occlusion, framing, motion blur, lighting and camera angle can change the landmark count or geometry. The 25% threshold is a simple evidence-based rule, not a calibrated probability or a claim that a sign intrinsically requires a particular number of hands.

Use the held-out figures as prototype measurements. For a stronger result, collect webcam examples in the intended environment, review the one-hand and two-hand false predictions, and evaluate on signers and sessions excluded from training.

Training status

The landmark MLP was trained for 17 epochs before early stopping. Its best validation-loss checkpoint reached 81.05% validation accuracy at epoch 11; the highest recorded validation accuracy was 83.78% at epoch 15. The deployed checkpoint now includes the hand-count support metadata used by the inference rule.